Sets and Functions

Last modified: 2026-10-13 18:25:48 (PDT)

Remark. Probability and statistics describe events, outcomes, and parameters as sets, and random variables, densities, and measures as functions. This page collects the definitions of sets and functions that the Morrison Lab's probability and statistics notes build on.

Sets

Definition 0.1 (Set)

Definition 0.1 (Set)**.** A **set** is a collection of distinct objects, called its elements. We write $a \in A$ when a is an element of the set A, and $a \notin A$ when it is not. Two sets are equal when they have exactly the same elements.

Remark 0.1 (Naive and formal set theory)

Remark 0.1 (Naive and formal set theory). Definition 0.1 is the informal ("naive") notion of a set, which is all these notes need. Formal set theory states axioms for sets instead (see Wikipedia: Set (mathematics)). For example, the last sentence of Definition 0.1, that two sets are equal when they have exactly the same elements, is one of those axioms, the *axiom of extensionality*.

Example 0.1 (Sets of die rolls)

Example 0.1 (Sets of die rolls)**.** The possible results of rolling a six-sided die form the set $\{1, 2, 3, 4, 5, 6\}$, with $3 \in \{1, 2, 3, 4, 5, 6\}$ and $7 \notin \{1, 2, 3, 4, 5, 6\}$. The sets $\{1, 2, 3\}$ and $\{3, 2, 1\}$ are equal, because they have the same elements; the order in which the elements are listed does not matter.

Definition 0.2 (Set-builder notation)

Definition 0.2 (Set-builder notation)**.** For a set A and a statement $P(x)$ about an element x, **set-builder notation** $\{x \in A : P(x)\}$ denotes the set of elements x of A for which $P(x)$ is true.

Example 0.2 (Even die rolls in set-builder notation)

Example 0.2 (Even die rolls in set-builder notation)**.** $\{x \in \{1,2,3,4,5,6\} : x \text{ is even}\} = \{2,4,6\}$ is the set of even rolls, and $\{x \in \mathbb{R} : x > 0\}$ is the set of positive real numbers.

Definition 0.3 (Interval)

Definition 0.3 (Interval)**.** For real numbers $a \leq b$, the **intervals** with **endpoints** a and b are:

- the **closed interval** $[a,b] \stackrel{\text{def}}{=} \{x \in \mathbb{R} : a \leq x \leq b\}$;
- the **open interval** $(a,b) \stackrel{\text{def}}{=} \{x \in \mathbb{R} : a < x < b\}$;
- the **half-open intervals** $[a,b) \stackrel{\text{def}}{=} \{x \in \mathbb{R} : a \leq x < b\}$ and $(a,b] \stackrel{\text{def}}{=} \{x \in \mathbb{R} : a < x \leq b\}$.

The **unbounded intervals** have at most one endpoint:

- $[a,\infty) \stackrel{\text{def}}{=} \{x \in \mathbb{R} : x \geq a\}$ and $(a,\infty) \stackrel{\text{def}}{=} \{x \in \mathbb{R} : x > a\}$;
- $(-\infty,b] \stackrel{\text{def}}{=} \{x \in \mathbb{R} : x \leq b\}$ and $(-\infty,b) \stackrel{\text{def}}{=} \{x \in \mathbb{R} : x < b\}$;
- $(-\infty,\infty) \stackrel{\text{def}}{=} \mathbb{R}$.

A square bracket means the endpoint belongs to the interval, and a parenthesis means it does not. The symbols ∞ and $-\infty$ are not real numbers, so they always get a parenthesis.

Example 0.3 (Which numbers are in an interval)

Example 0.3 (Which numbers are in an interval)**.**

- $1 \in [1,2]$, but $1 \notin (1,2)$.
- $2 \in (1,2]$, but $2 \notin [1,2)$.
- 1.5 is in all four of $[1,2]$, $(1,2)$, $[1,2)$, and $(1,2]$.
- $[0,\infty)$ is the set of non-negative real numbers: $0 \in [0,\infty)$ and $1000 \in [0,\infty)$, but $-0.1 \notin [0,\infty)$.

Definition 0.4 (Subset)

Definition 0.4 (Subset)**.** A set A is a **subset** of a set B, written $A \subseteq B$, if every element of A is an element of B.

Example 0.4 (Even die rolls)

Example 0.4 (Even die rolls)**.** The even die rolls form a subset of the possible rolls: $\{2, 4, 6\} \subseteq \{1, 2, 3, 4, 5, 6\}$, because each of 2, 4, and 6 is a possible roll. The set $\{2, 4, 7\}$ is not a subset of $\{1, 2, 3, 4, 5, 6\}$, because 7 is not a possible roll.

Theorem 0.1 (Every set is a subset of itself)

Theorem 0.1 (Every set is a subset of itself)**.** *For every set A, $A \subseteq A$.*

Proof

Proof. Every element of A is an element of A, which is Definition 0.4 with $B = A$. □

Definition 0.5 (Strict subset)

Definition 0.5 (Strict subset)**.** A set A is a **strict subset** of a set B, written $A \subsetneq B$, if A is a subset of B and $A \neq B$.

Remark 0.2 (Proper subsets and the symbol $\subset$)

Remark 0.2 (Proper subsets and the symbol $\subset$). Other sources call a strict subset a **proper subset**. Sources disagree about the symbol $\subset$: some use it for "subset" ($\subseteq$), and others for "strict subset" ($\subsetneq$) (see Wikipedia: Subset). For example, $\{1, 2\} \subset \{1, 2\}$ is true in the first reading, because $\{1, 2\} \subseteq \{1, 2\}$, and false in the second, because the two sets are equal. These notes avoid $\subset$ and write $\subseteq$ or $\subsetneq$.

Definition 0.6 (Superset)

Definition 0.6 (Superset)**.** A set B is a **superset** of a set A, written $B \supseteq A$, if every element of A is an element of B.

Remark 0.3 (Superset and subset)

Remark 0.3 (Superset and subset). $B \supseteq A$ means the same thing as $A \subseteq B$ (subset). For example, $\{1, 2, 3\} \supseteq \{1, 3\}$, and $\{1, 3\} \subseteq \{1, 2, 3\}$. But $\{1, 3\}$ is not a superset of $\{1, 2\}$,

because $2 \in \{1, 2\}$ and $2 \notin \{1, 3\}$.

Definition 0.7 (Strict superset)

Definition 0.7 (Strict superset)**.** A set B is a **strict superset** of a set A, written $B \supsetneq A$, if B is a superset of A and $B \neq A$.

Remark 0.4 (Strict superset and strict subset)

Remark 0.4 (Strict superset and strict subset). $B \supsetneq A$ means the same thing as $A \subsetneq B$ (strict subset). For example, $\{1, 2, 3\} \supsetneq \{1, 3\}$, because $\{1, 2, 3\}$ contains 2 and $\{1, 3\}$ does not.

Example 0.5 (Strict subsets and supersets of die rolls)

Example 0.5 (Strict subsets and supersets of die rolls)**.** For the even rolls $A = \{2, 4, 6\}$ and the possible rolls $B = \{1, 2, 3, 4, 5, 6\}$:

- $A \subsetneq B$, because $A \subseteq B$ (Example 0.4) and $1 \in B$ but $1 \notin A$, so $A \neq B$;
- $B \supseteq A$ and $B \supsetneq A$, for the same reasons;
- $B \subseteq B$, but B is not a strict subset of itself, because $B = B$; likewise $B \supseteq B$, but B is not a strict superset of itself, because $B = B$.

Definition 0.8 (Empty set)

Definition 0.8 (Empty set)**.** The **empty set**, denoted $\emptyset$, is the set that has no elements.

Remark 0.5 (Other notation for the empty set)

Remark 0.5 (Other notation for the empty set). Other sources write $\{\}$ for $\emptyset$ (see Wikipedia: Empty set). Some sources call it the **null set**, but in measure theory a "null set" usually means a set of measure zero, which need not be empty. For example, the interval $[0, 0] = \{0\}$ has length $0 - 0 = 0$, so it is a null set in that sense, but it is not empty, because $0 \in \{0\}$.

Example 0.6 (Impossible die rolls)

Example 0.6 (Impossible die rolls)**.** No roll of a six-sided die is greater than 6, so $\{x \in \{1, 2, 3, 4, 5, 6\} : x > 6\} = \emptyset$.

Definition 0.9 (Vacuous truth)

Definition 0.9 (Vacuous truth)**.** A statement of the form "every element of A has property P" is **vacuously true** when A is the empty set: A has no elements, so no element of A fails to have property P.

Example 0.7 (Every impossible die roll is even)

Example 0.7 (Every impossible die roll is even)**.** The set of die rolls greater than 6 is $\emptyset$ (Example 0.6). So "every die roll greater than 6 is even" is vacuously true, and so is "every die roll greater than 6 is odd": there is no such roll to fail either statement.

Theorem 0.2 (There is only one empty set)

Theorem 0.2 (There is only one empty set)**.** *If E_1 and E_2 are sets with no elements, then $E_1 = E_2$.*

Proof

Proof. Neither set has any elements, so every element of E_1 is an element of E_2 and vice versa, vacuously. Sets with the same elements are equal (Definition 0.1). □

Theorem 0.3 (The empty set is a subset of every set)

Theorem 0.3 (The empty set is a subset of every set)**.** *For every set A, $\emptyset \subseteq A$.*

Proof

Proof. By Definition 0.4, $\emptyset \subseteq A$ fails only if some element of $\emptyset$ is not in A. The empty set has no elements, so no such element exists. □

Definition 0.10 (Collection of sets (family of sets))

Definition 0.10 (Collection of sets (family of sets))**.** A **collection of sets** (also called a **family of sets**) is a set whose elements are themselves sets.

Example 0.8 (The collection of all subsets of a two-element set)

Example 0.8 (The collection of all subsets of a two-element set)**.** The subsets of $\{1, 2\}$ are $\emptyset$, $\{1\}$, $\{2\}$, and $\{1, 2\}$ (Theorem 0.3 and Theorem 0.1 give the first and the last), so

the collection of all subsets of $\{1, 2\}$ is

$$\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

It has four elements, each of them a set. The collection $\{\emptyset\}$ has one element, the empty set, so it is not equal to $\emptyset$, which has none.

Combining sets

Definition 0.11 (Union)

Definition 0.11 (Union)**.** The **union** of sets A and B, written $A \cup B$, is the set of elements that are in A, in B, or in both:

$$A \cup B \stackrel{\text{def}}{=} \{x : x \in A \text{ or } x \in B\}$$

Definition 0.12 (Union of a family of sets)

Definition 0.12 (Union of a family of sets)**.** Let I be a set of indices, such as $\{1, \dots, n\}$ or $\{1, 2, 3, \dots\}$, and let A_i be a set for each $i \in I$. The **union** of the sets A_i, written $\bigcup_{i \in I} A_i$, is the set of elements that are in at least one A_i. For $I = \{1, \dots, n\}$, it is also written $\bigcup_{i=1}^{n} A_i$ or $A_1 \cup \cdots \cup A_n$, and for $I = \{1, 2, 3, \dots\}$, it is written $\bigcup_{i=1}^{\infty} A_i$.

Definition 0.13 (Intersection)

Definition 0.13 (Intersection)**.** The **intersection** of sets A and B, written $A \cap B$, is the set of elements that are in both A and B:

$$A \cap B \stackrel{\text{def}}{=} \{x : x \in A \text{ and } x \in B\}$$

Definition 0.14 (Intersection of a family of sets)

Definition 0.14 (Intersection of a family of sets)**.** Let I be a nonempty set of indices, such as $\{1, \dots, n\}$ or $\{1, 2, 3, \dots\}$, and let A_i be a set for each $i \in I$. The **intersection** of the sets A_i, written $\bigcap_{i \in I} A_i$, is the set of elements that are in every A_i. For $I = \{1, \dots, n\}$, it is also written $\bigcap_{i=1}^{n} A_i$ or $A_1 \cap \cdots \cap A_n$, and for $I = \{1, 2, 3, \dots\}$, it is written $\bigcap_{i=1}^{\infty} A_i$.

Definition 0.15 (Set difference)

Definition 0.15 (Set difference)**.** The **set difference** of sets A and B, written $A \setminus B$, is the set of elements of A that are not in B:

$$A \setminus B \stackrel{\text{def}}{=} \{x \in A : x \notin B\}$$

Definition 0.16 (Complement)

Definition 0.16 (Complement)**.** Let A be a subset of a set S. The **complement** of A in S is the set difference (Definition 0.15) $S \setminus A$: the elements of S that are not in A. When the set S is clear from context, the complement of A is written A^c:

$$A^c \stackrel{\text{def}}{=} S \setminus A = \{x \in S : x \notin A\}$$

Example 0.9 (Complements of sets of die rolls)

Example 0.9 (Complements of sets of die rolls)**.** Let $S = \{1, 2, 3, 4, 5, 6\}$ be the possible rolls of a die.

- The complement of the even rolls $A = \{2, 4, 6\}$ is $A^c = \{1, 3, 5\}$, the odd rolls.
- The complement of S itself is $S^c = \emptyset$, and the complement of $\emptyset$ is $\emptyset^c = S$.

The complement depends on S: in $S = \{1, 2, 3, 4, 5, 6\}$, the complement of $\{2, 4, 6\}$ is $\{1, 3, 5\}$, but in $S = \{2, 4, 6, 8\}$, it is $\{8\}$.

Example 0.10 (Combining sets of die rolls)

Example 0.10 (Combining sets of die rolls)**.** Let $A = \{2, 4, 6\}$ (the even rolls) and $B = \{1, 2, 3\}$ (the rolls of at most 3). Then:

- $A \cup B = \{1, 2, 3, 4, 6\}$, the rolls that are even or at most 3;
- $A \cap B = \{2\}$, the only roll that is both;
- $A \setminus B = \{4, 6\}$, the even rolls greater than 3;
- $\{2, 4, 6\} \cap \{1, 3, 5\} = \emptyset$: no roll is both even and odd.

Example 0.11 (Unions and intersections of many sets)

Example 0.11 (Unions and intersections of many sets)**.**

- $$\bigcup_{i=1}^{3}\{i, i+1\} = \{1,2\} \cup \{2,3\} \cup \{3,4\}$$
$$= \{1,2,3,4\}.$$

- $\bigcup_{i=1}^{\infty}\{i\} = \mathbb{N}$: each natural number n is in the set $\{n\}$, and each set $\{i\}$ contains only the natural number i.
- $\bigcap_{i=1}^{\infty}[0, 1/i] = \{0\}$: 0 is in every interval $[0, 1/i]$; a number $x > 0$ is not in $[0, 1/i]$ once $i > 1/x$; and a number $x < 0$ is in none of them.

Theorem 0.4 (De Morgan's laws)

Theorem 0.4 (De Morgan's laws)**.** *Let S be a set, let I be a nonempty set of indices, and let A_i be a subset of S for each $i \in I$. Then the complement of a union is the intersection of the complements, and the complement of an intersection is the union of the complements:*

$$S \setminus \bigcup_{i\in I} A_i = \bigcap_{i\in I}(S \setminus A_i)$$
$$S \setminus \bigcap_{i\in I} A_i = \bigcup_{i\in I}(S \setminus A_i)$$

With the complement notation of Definition 0.16, these read $\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c$ and $\left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c$.

Proof

Proof. Two sets are equal when they have the same elements (Definition 0.1), so it is enough to show that an element x is in the left-hand side if and only if ($\iff$) it is in the right-hand side. For the first law:

$$\begin{aligned}
x \in S \setminus \bigcup_{i\in I} A_i &\iff x \in S \text{ and } x \notin \bigcup_{i\in I} A_i && \text{(Definition 0.15)}\\
&\iff x \in S \text{ and } x \notin A_i \text{ for every } i \in I && \text{(Definition 0.11)}\\
&\iff x \in S \setminus A_i \text{ for every } i \in I && \text{(Definition 0.15)}\\
&\iff x \in \bigcap_{i\in I}(S \setminus A_i) && \text{(Definition 0.13)}
\end{aligned}$$

For the second law:

$$
\begin{aligned}
x \in S \setminus \bigcap_{i \in I} A_i &\iff x \in S \text{ and } x \notin \bigcap_{i \in I} A_i && \text{(Definition 0.15)} \\
&\iff x \in S \text{ and } x \notin A_i \text{ for some } i \in I && \text{(Definition 0.13)} \\
&\iff x \in S \setminus A_i \text{ for some } i \in I && \text{(Definition 0.15)} \\
&\iff x \in \bigcup_{i \in I} (S \setminus A_i) && \text{(Definition 0.11)}
\end{aligned}
$$

□

Example 0.12 (De Morgan's laws for die rolls)

Example 0.12 (De Morgan's laws for die rolls)**.** In $S = \{1, 2, 3, 4, 5, 6\}$, let $A = \{2, 4, 6\}$ and $B = \{1, 2, 3\}$, as in Example 0.10, so $A^c = \{1, 3, 5\}$ and $B^c = \{4, 5, 6\}$.

- $$
  \begin{aligned}
  (A \cup B)^c &= \{1, 2, 3, 4, 6\}^c \\
  &= \{5\},
  \end{aligned}
  $$

  and

  $$
  \begin{aligned}
  A^c \cap B^c &= \{1, 3, 5\} \cap \{4, 5, 6\} \\
  &= \{5\}.
  \end{aligned}
  $$

- $$
  \begin{aligned}
  (A \cap B)^c &= \{2\}^c \\
  &= \{1, 3, 4, 5, 6\},
  \end{aligned}
  $$

  and

  $$
  \begin{aligned}
  A^c \cup B^c &= \{1, 3, 5\} \cup \{4, 5, 6\} \\
  &= \{1, 3, 4, 5, 6\}.
  \end{aligned}
  $$

Definition 0.17 (Ordered pair)

Definition 0.17 (Ordered pair)**.** An **ordered pair** (a, b) is a pair of objects listed in order, with a first and b second; two ordered pairs are equal, $(a, b) = (c, d)$, when $a = c$ and $b = d$.

Definition 0.18 (Cartesian product)

Definition 0.18 (Cartesian product)**.** The **Cartesian product** of sets A and B, written $A \times B$, is the set of ordered pairs whose first entry is in A and whose second entry is in B:

$$A \times B \stackrel{\text{def}}{=} \{(a, b) : a \in A \text{ and } b \in B\}$$

Example 0.13 (Cartesian products)

Example 0.13 (Cartesian products)**.**

- $\{1, 2\} \times \{5, 6\} = \{(1, 5), (1, 6), (2, 5), (2, 6)\}$. The pair $(5, 1)$ is not in it, because its first entry, 5, is not in $\{1, 2\}$; so $(1, 5) \neq (5, 1)$.
- The results of rolling a die twice form $\{1, \ldots, 6\} \times \{1, \ldots, 6\}$, which has $6 \cdot 6 = 36$ elements, such as $(3, 5)$: a 3 on the first roll and a 5 on the second.
- $[0, 2] \times [0, 3]$ is the set of points (x, y) in the plane with $0 \leq x \leq 2$ and $0 \leq y \leq 3$: a rectangle of width 2 and height 3.

Functions

Definition 0.19 (Function (map, mapping, transformation))

Definition 0.19 (Function (map, mapping, transformation))**.** A **function** f from a set A to a set B, written $f : A \to B$, assigns to each element $a \in A$ exactly one element $f(a) \in B$.

- The element $a \in A$ is called the **argument** (or **input**) to f.
- The assigned element $f(a) \in B$ is called the **value** (or **output**) of f at a.

A function is also called a **map** (or **mapping**) or a **transformation**; on this site the three words mean the same thing, although some authors reserve "transformation" for a function from a set to itself.

Example 0.14 (Doubling a die roll)

Example 0.14 (Doubling a die roll)**.** The rule $f(a) = 2a$ defines a function $f : \{1, 2, 3, 4, 5, 6\} \to \mathbb{R}$, because it assigns exactly one real number to each possible roll; for example, $f(3) = 6$. A rule that assigned both 2 and -2 to the roll 1 would not be a function, because a function assigns exactly one value to each element.

Definition 0.20 (Relation)

Definition 0.20 (Relation)**.** A **relation** $\sim$ on a set S is a rule that, for each ordered pair of elements $a, b \in S$, says whether $a \sim b$ holds ("a is related to b").

Example 0.15 (Relations on the real numbers)

Example 0.15 (Relations on the real numbers)**.** $=$, $<$ and $\leq$ are relations on $\mathbb{R}$. For $<$, the pair $(2, 5)$ is related, since $2 < 5$, but the pair $(5, 2)$ is not, since $5 < 2$ is false: the order of the pair matters.

Definition 0.21 (Domain)

Definition 0.21 (Domain)**.** The **domain** of a function $f : A \to B$ is the set A of elements that f assigns values to.

Definition 0.22 (Codomain)

Definition 0.22 (Codomain)**.** The **codomain** of a function $f : A \to B$ is the set B that its values are required to lie in.

Definition 0.23 (Image)

Definition 0.23 (Image)**.** The **image** of a function $f : A \to B$ is the set of values that f actually takes:

$$f(A) \stackrel{\text{def}}{=} \{f(a) : a \in A\}$$

For a single element $a \in A$, the value $f(a)$ is also called the **image of** a under f.

Remark 0.6 (Image and range)

Remark 0.6 (Image and range)*.* Many sources call the image the **range**, but others use "range" for the codomain, so these notes avoid "range" for functions in general (see Wikipedia: Range of a function). For example, for $f : \mathbb{R} \to \mathbb{R}$ with $f(a) = a^2$, "the range of f" could mean the image $[0, \infty)$ or the codomain $\mathbb{R}$.
The image is always a subset of the codomain, because every value $f(a)$ lies in B.

Example 0.16 (Domain, codomain, and image of the doubled die roll)

Example 0.16 (Domain, codomain, and image of the doubled die roll)**.** For the function $f : \{1, 2, 3, 4, 5, 6\} \to \mathbb{R}$ with $f(a) = 2a$ from Example 0.14:

- the domain is $\{1, 2, 3, 4, 5, 6\}$;
- the codomain is $\mathbb{R}$;
- the image is $\{2, 4, 6, 8, 10, 12\}$.

The image is a subset of the codomain, and here a much smaller one: most real numbers, such as 3 and -1, are not values of f.

Definition 0.24 (Graph of a function)

Definition 0.24 (Graph of a function)**.** The **graph** of a function $f : A \to B$ is the set of ordered pairs $(a, f(a))$, one for each element a of its domain:

$$\{(a, f(a)) : a \in A\}$$

Example 0.17 (The graph of a function on three points)

Example 0.17 (The graph of a function on three points)**.** Let $f : \{-1, 0, 2\} \to \mathbb{R}$ with $f(a) = a^2$. Then $f(-1) = 1$, $f(0) = 0$, and $f(2) = 4$, so the graph of f is $\{(-1, 1), (0, 0), (2, 4)\}$. The pair $(1, 1)$ is not in the graph, because 1 is not in the domain of f.

Remark 0.7 (The graph of a real function is a curve)

Remark 0.7 (The graph of a real function is a curve). For a function $f : \mathbb{R} \to \mathbb{R}$, the graph is a curve in the plane, with a on the horizontal axis and $f(a)$ on the vertical axis.

Remark 0.8 (What a function is fundamentally)

Remark 0.8 (What a function is fundamentally). Definition 0.19 defines what a function *does*: it assigns an output to each input. In formal set theory, what a function *is* is described in terms of ordered pairs and its graph (Definition 0.24): the set of pairs $G = \{(a, f(a)) : a \in A\} \subseteq A \times B$. Authors define a function formally in one of two ways:

1. **As its graph** (Halmos 1974, sec. 8, p. 30): a function is a subset of $A \times B$ containing exactly one pair (a, b) for each $a \in A$, where $b = f(a)$, and the statement $f(a) = b$ is notation for $(a, b) \in f$. Under this convention, the codomain B is specified alongside the graph in the declaration $f : A \to B$.
2. **As a triple** (A, B, G): the function consists of the domain A, the codomain B, and the graph $G \subseteq A \times B$. This explicit packaging retains the codomain B as part of the function itself, distinguishing functions that have the same graph but different codomains (such as a function mapping into $\mathbb{R}$ versus into integers).

For example, the die-doubling function in Example 0.14 has the graph of six ordered pairs $\{(1, 2), (2, 4), (3, 6), (4, 8), (5, 10), (6, 12)\}$.
This distinction clarifies what a function is and is not:

- **Not an algebraic formula:** A function is not fundamentally an equation or algebraic expression. While many functions are specified by a formula (such as $f(x) = x^2$), many others cannot be written in closed form (such as empirical measurements or a lookup table), yet every function has a well-defined set of input-output pairs.
- **Not a curve:** A curve or plot is a geometric visualization of a function whose domain and codomain are subsets of $\mathbb{R}$, but functions can map between arbitrary sets (such as sets of words, matrices, or probability distributions).
- **Sets of pairs with unique first entries:** Any subset of $A \times B$ in which no two distinct pairs share the same first element defines a function from the set of those first elements to B.

In computer science, viewing a function as a set of key-value pairs corresponds directly to a **dictionary**, **lookup table**, or **hash map**. Evaluating an expression computes outputs on demand; caching computed outputs $(a, f(a))$ on demand to return on future calls without recomputing is known as **memoization**.

Definition 0.25 (Composition)

Definition 0.25 (Composition)**.** Let $g : A \to B$ and $f : B \to C$ be functions. The **composition** of f and g is the function $f \circ g : A \to C$ defined by

$$(f \circ g)(a) \stackrel{\text{def}}{=} f(g(a)) \quad \text{for each } a \in A.$$

In $f \circ g$, g is the **inner function**, applied first, and f is the **outer function**, applied to the result.

Example 0.18 (The order of composition matters)

Example 0.18 (The order of composition matters)**.** Let $g : \mathbb{R} \to \mathbb{R}$ with $g(x) = x + 1$, and $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$.

- $$\begin{aligned}(f \circ g)(2) &= f(g(2)) \\ &= f(3) \\ &= 9 :\end{aligned}$$

  g is the inner function, and f is the outer function.

- $$\begin{aligned}(g \circ f)(2) &= g(f(2)) \\ &= g(4) \\ &= 5 :\end{aligned}$$

now f is the inner function, and g is the outer function.

Since $9 \neq 5$, $f \circ g$ and $g \circ f$ are different functions.

Definition 0.26 (Inverse function)

Definition 0.26 (Inverse function)**.** Let $f : A \to B$ be a function. A function $g : B \to A$ is the **inverse function** of f, written f^{-1}, if both

- $g(f(a)) = a$ for every $a \in A$, and
- $f(g(b)) = b$ for every $b \in B$.

A function that has an inverse function is **invertible**.

Remark 0.9 (f^{-1} is not $1/f$)

Remark 0.9 (f^{-1} is not $1/f$)*.* The -1 in f^{-1} does not mean a reciprocal: $f^{-1}(b)$ is the input that f maps to b, not $\frac{1}{f(b)}$. For example, with $f(x) = 2x$ on $\mathbb{R}$, $f^{-1}(4) = 2$, but $\frac{1}{f(4)} = \frac{1}{8}$.

Example 0.19 (An invertible function, and one that is not)

Example 0.19 (An invertible function, and one that is not)**.**

- Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = 2x + 1$. Its inverse function is $f^{-1}(y) = \frac{y-1}{2}$. For example, $f(3) = 7$ and

$$\begin{aligned} f^{-1}(7) &= \frac{7-1}{2} \\ &= 3. \end{aligned}$$

- Let $h : \mathbb{R} \to \mathbb{R}$ with $h(x) = x^2$. Then $h(-2) = 4$ and $h(2) = 4$, so an inverse function would need $h^{-1}(4)$ to equal both -2 and 2. So h is not invertible.

Definition 0.27 (Elementwise application of a function to a vector)

Definition 0.27 (Elementwise application of a function to a vector)**.** Let $f : A \to B$ be a function (definition), and let $\tilde{x} = (x_1, x_2, \ldots, x_n)^\top$ be a vector whose entries all lie in A. The **elementwise application** of f to $\tilde{x}$, written $f^{\odot}(\tilde{x})$, is the vector of the values of f at the entries of $\tilde{x}$:

$$f^{\odot}(\tilde{x}) = (f(x_1), f(x_2), \dots, f(x_n))^{\top}.$$

It is a function $f^{\odot} : A^n \to B^n$, where A^n is the n-fold Cartesian product of A with itself (definition).

Example 0.20 (The exponential and logistic functions applied to a vector)

Example 0.20 (The exponential and logistic functions applied to a vector)**.** Let $\tilde{x} = (-1, 0, 2)^{\top}$. With $\exp(t) = e^t$ and the logistic function $\operatorname{expit}(t) = \frac{1}{1+e^{-t}}$,

$$\exp^{\odot}(\tilde{x}) = (e^{-1}, e^0, e^2)^{\top},$$
$$\operatorname{expit}^{\odot}(\tilde{x}) = (\operatorname{expit}(-1), \operatorname{expit}(0), \operatorname{expit}(2))^{\top}.$$

The code evaluates both, using R's `exp()` and `plogis()`, which apply to a vector entry by entry.

```
x <- c(-1, 0, 2)
exp_x <- exp(x)
expit_x <- plogis(x)
exp_x
#> [1] 0.3678794 1.0000000 7.3890561
expit_x
#> [1] 0.2689414 0.5000000 0.8807971
```

So $\exp^{\odot}(\tilde{x}) \approx (0.3679, 1, 7.3891)$ and $\operatorname{expit}^{\odot}(\tilde{x}) \approx (0.2689, 0.5, 0.8808)$. Each entry of the output depends only on the entry of $\tilde{x}$ in the same position.

Remark 0.10 (Notation and software for elementwise application)

Remark 0.10 (Notation and software for elementwise application).

- The superscript $\odot$ keeps $f^{\odot}$ apart from f itself. Where a function f takes a whole vector as its argument, as in the multivariable calculus of vector calculus, $f(\tilde{x})$ means f evaluated at the vector, which is generally not the vector of its values at the entries.
- If f has parameters, as in $f_{\tilde{\beta}}$, the same parameters are used at every entry.
- Software usually applies a scalar function to a vector this way, without a special symbol: R and NumPy evaluate `exp(x)` entry by entry. The matrix exponential of a square matrix is a different function from $\exp^{\odot}$.

Definition 0.28 (Sequence)

Definition 0.28 (Sequence)**.** A **sequence** in a set S is a function $a : \mathbb{N} \to S$ from the natural numbers to S. Its value $a(n)$ is written a_n and called its nth **term**, and the sequence is written (a_n) or $a_1, a_2, a_3, ...$

Example 0.21 (Sequences)

Example 0.21 (Sequences)**.**

- $a_n = \frac{1}{n}$ is the sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, ...$; its third term is $a_3 = \frac{1}{3}$.
- $b_n = (-1)^n$ is the sequence $-1, 1, -1, 1, ...$: a sequence can repeat values.
- The die rolls $4, 1, 4$ form a finite sequence of length 3, with $a_1 = 4$, $a_2 = 1$, and $a_3 = 4$. As a set, $\{4, 1, 4\} = \{1, 4\}$, but as a sequence, the order and the repeats matter.

Definition 0.29 (Finite sequence)

Definition 0.29 (Finite sequence)**.** Let S be a set. A **finite sequence** of length n is a function from $\{1, \dots, n\}$ to S, written $a_1, \dots, a_n$.

Definition 0.30 (Expression)

Definition 0.30 (Expression)**.** An **expression** is a string of symbols, built from constants, variables, operations and function names according to the rules of the notation. When its variables are read as inputs, an expression *denotes* a function: it fixes the output for each input.

Example 0.22 (Two expressions for one function)

Example 0.22 (Two expressions for one function)**.** The expressions $ax^2 + bx + c$ and $(ax + b)\,x + c$ denote the same function of x for any fixed a, b and c, because they give the same value at every x.
Precedence parses $ax^2 + bx + c$ as a sum of three terms: the power x^2 is applied first, then a multiplies x^2 and b multiplies x, and the three products are added. The parentheses in $(ax + b)\,x + c$ force a different tree: a multiplies x, b is added, the sum multiplies x, and c is added.

Definition 0.31 (Parse tree)

Definition 0.31 (Parse tree)**.** The **parse tree** of an expression is the tree that shows

how its symbols group into operations. Its leaves are the constants and variables of the expression, and each internal node is an operation applied to the nodes below it.

Example 0.23 (The parse tree of a polynomial)

Example 0.23 (The parse tree of a polynomial)**.** The parse tree of $(ax + b)\,x + c$, with each operation having two children, is:

- the root $+$, whose children are
  - the node $\times$, whose children are
    - the node $+$, whose children are
      - the node $\times$, whose children are the leaves a and x,
      - the leaf b;
    - the leaf x;
  - the leaf c.

Read from the leaves up, the tree says to multiply a by x, add b, multiply by x, and add c. The expression $ax^2 + bx + c$ has a different tree. Its root is the $+$ whose children are $ax^2 + bx$ and c, and the first of these is a $+$ whose children are ax^2 and bx.

Remark 0.11 (Expressions, precedence and parse trees)

Remark 0.11 (Expressions, precedence and parse trees). An expression does not say how to compute the output it fixes, and many expressions denote the same function. A function is the input-output relationship itself; an expression is one way of writing it down.
The rules of operator precedence (PEMDAS: parentheses, exponents, multiplication and division, addition and subtraction) decide how the symbols group. They turn the string into its parse tree (Definition 0.31).

Definition 0.32 (Algorithm)

Definition 0.32 (Algorithm)**.** An **algorithm** is a finite list of unambiguous steps that, applied to any valid input, halts and returns an output. An algorithm *computes* a function: for each input it returns the function's value at that input.

Example 0.24 (Two algorithms for one function)

Example 0.24 (Two algorithms for one function)**.** Take the function that Example 0.22 writes two ways. Evaluating the tree of $ax^2 + bx + c$, with x^2 computed as $x \cdot x$, takes three multiplications and two additions. Evaluating the tree of Horner's form $(ax + b)\,x + c$

takes two multiplications and two additions. At $a = 2$, $b = -3$, $c = 5$ and $x = 1.5$, the first algorithm computes $4.5 - 4.5 + 5 = 5$ and the second computes $(3 - 3) \cdot 1.5 + 5 = 5$, the same value by different steps.

Remark 0.12 (How functions, expressions and algorithms relate)

Remark 0.12 (How functions, expressions and algorithms relate). The three ideas fit together this way:

- A function is the input-output relationship.
- An expression denotes a function, and many expressions denote the same one.
- An algorithm computes a function, and many algorithms compute the same one.

An expression is not an algorithm, because it does not say what to do, or in what order. Given values for its variables, evaluating its parse tree from the leaves up is one algorithm for the function it denotes, and a different expression for the same function gives a different tree and a different algorithm.

Remark 0.13 (Values with no known algorithm)

Remark 0.13 (Values with no known algorithm). Some functions have no known finite algorithm for their exact values, and expressions that denote them, such as a limit, a supremum, an integral with no closed form, or an $\arg\min$ over a set with no formula for the minimizer, cannot be evaluated by following their tree. Such a value can still be well defined and approximated by an algorithm.

Example 0.25 (A minimizer with no known formula)

Example 0.25 (A minimizer with no known formula)**.** The minimizer $\arg\min_{x \in \mathbb{R}} (x - \cos x)^2$ is well defined, because $x - \cos x$ is strictly increasing and equals zero at one point, yet that point has no known closed-form expression, so no known finite list of arithmetic steps returns it exactly. An algorithm such as bisection approximates it as closely as we like.

Countable sets

Definition 0.33 (Finite set and cardinality)

Definition 0.33 (Finite set and cardinality)**.** A set A is **finite** if it is empty, or if, for some natural number n, its elements can be listed as a finite sequence (Definition 0.29) $a_1, \ldots, a_n$ in which each element of A appears exactly once. That number n is the **cardinality** of

A, its number of elements, written $|A|$; the empty set has cardinality $|\emptyset| = 0$.

Remark 0.14 (Two meanings of $|\cdot|$)

Remark 0.14 (Two meanings of $|\cdot|$). Applied to a set, the bars $|A|$ mean the cardinality of A. Applied to a number, $|x|$ means its absolute value, or its modulus when x is complex. For example, $|\{-3\}| = 1$, but $|-3| = 3$.

Example 0.26 (Finite and infinite sets)

Example 0.26 (Finite and infinite sets)**.**

- $\{2, 4, 6\}$ is finite, with $|\{2, 4, 6\}| = 3$.
- $\{1, 1, 2\} = \{1, 2\}$, so $|\{1, 1, 2\}| = 2$: a repeated listing does not count twice.
- $\emptyset$ is finite, with $|\emptyset| = 0$.
- The natural numbers $\mathbb{N} = \{1, 2, 3, ...\}$ are infinite: for any n, a list $a_1, ..., a_n$ of natural numbers misses the natural number $a_1 + \cdots + a_n + 1$, which is larger than every number in the list.

Definition 0.34 (Infinite set)

Definition 0.34 (Infinite set)**.** A set that is not finite (Definition 0.33) is **infinite**.

Definition 0.35 (Countable set)

Definition 0.35 (Countable set)**.** A set is **countable** if it is finite (Definition 0.33), or if its elements can be listed as a sequence (Definition 0.28) $a_1, a_2, a_3, ...$ in which every element appears.

Definition 0.36 (Countably infinite set)

Definition 0.36 (Countably infinite set)**.** A set is **countably infinite** if it is countable and not finite.

Example 0.27 (Countable and uncountable sets)

Example 0.27 (Countable and uncountable sets)**.**

- $\{1, 2, 3, 4, 5, 6\}$ is countable, because it is finite.
- The non-negative integers $\{0, 1, 2, ...\}$ are countably infinite: the sequence $0, 1, 2, ...$ lists them.

- The integers are countably infinite too: the sequence $0, 1, -1, 2, -2, 3, -3, \ldots$ lists every one of them.

Theorem 0.5 (The unit interval is not countable)

Theorem 0.5 (The unit interval is not countable)**.** *The interval* $[0, 1]$ *is not countable.*

Remark 0.15 (Cantor's diagonal argument)

Remark 0.15 (Cantor's diagonal argument). The standard proof is Cantor's diagonal argument: given any sequence of numbers in $[0, 1]$, it builds a number in $[0, 1]$ whose decimal expansion differs from the nth number's in the nth digit (its nth decimal place, one of $0, 1, \ldots, 9$), so the sequence misses it (see Wikipedia: Cantor's diagonal argument). One way to choose the nth digit is to use 5, unless the nth number's nth digit is 5, in which case use 4. For example, if the sequence starts $0.1234\ldots$, $0.3579\ldots$, $0.2468\ldots$, the diagonal digits are 1, 5, and 6, so the new number starts $0.545\ldots$, which differs from the first number in the first digit, from the second in the second digit, and from the third in the third digit.

Extended non-negative real numbers

Definition 0.37 (Extended non-negative real numbers)

Definition 0.37 (Extended non-negative real numbers)**.** The **extended non-negative real numbers**, written $[0, \infty]$, are the non-negative real numbers together with an extra element ∞ that is greater than every real number:

$$[0, \infty] \stackrel{\text{def}}{=} [0, \infty) \cup \{\infty\}$$

Addition extends to $[0, \infty]$ by setting

$$\begin{aligned} x + \infty &= \infty + x \\ &= \infty \end{aligned}$$

for every $x \in [0, \infty]$.

Remark 0.16 (Arithmetic with ∞)

Remark 0.16 (Arithmetic with ∞). ∞ is not a real number, so some real-number arithmetic does not extend to it: $\infty - \infty$ is left undefined, and $\infty + 1 = \infty + 2$ does not imply $1 = 2$. Measure theory uses $[0, \infty]$ because the size of a set, such as the length of the whole real

line, can be infinite (see Wikipedia: Extended real number line).

Example 0.28 (Adding with ∞)

Example 0.28 (Adding with ∞)**.** In $[0, \infty]$, $2 + 3 = 5$ as usual, while $2 + \infty = \infty$ and $\infty + \infty = \infty$.

Further reading

These books treat sets and the logic behind them in more depth.

- Devlin (1993) is an undergraduate introduction to axiomatic set theory. It covers the axioms of set theory, ordinals, and cardinals, which go well beyond the countable sets and functions on this page.
- Halmos (1974) is a short, informal treatment of the same axioms, covering relations, functions, families of sets, and cardinal numbers.
- Enderton (2001) is a standard introduction to mathematical logic. Its first chapter collects the facts about sets, relations, and functions that the later chapters use.
- Barker-Plummer et al. (2011) teaches propositional and first-order logic, the language behind statements such as "for every set A there exists…".

References

Barker-Plummer, Dave, Jon Barwise, and John Etchemendy. 2011. *Language, Proof and Logic*. 2nd ed. CSLI Publications. https://www.amazon.com/dp/1575866323.

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